



CANTINAIQ IN PRACTICE.

An evaluation of the Slurpini partner-intelligence case from the perspective of reproducible data products, governance, and the cost of doing more than the brief asked.



**From notebook
to data product.**

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A BRIEF ON CANTINAIQ IN PRACTICE

Introduction.

The Slurpini case asks three things: extend the Vivino crawler, perform exploratory analysis, deliver an evidence-based recommendation. That is the brief. This document evaluates what changes when those three deliverables are answered as a reproducible data product instead of a one-off analysis.

I answered the brief twice. The first track, **/bare**, is the assignment in roughly 150 lines of code: one pandas notebook, a first-word producer heuristic, a half-page recommendation. It runs in thirty seconds and meets the brief honestly, including listing its own limitations.

The second track, **/supercharged**, treats the same business question as a data product. Polars and DuckDB for processing, Pandera schema contracts at every stage boundary, a five-factor Bayesian composite score, an OpenRouter-driven LLM disambiguation pass for producer names, property-based tests on the scoring math, Jinja-templated reports whose numbers come from a config-hashed run log, a Vite dashboard, and external-validity work that quantifies Vivino bias against Italian Trade Agency NL imports.

This document is not a summary of either track. It is an evaluation of when a heavyweight architecture earns its place against the next-best alternative — usually a notebook — and where the line between rigour and over-engineering actually sits for a final assignment. The course material understandably assumes you will pick one. The brief asks for one. The interesting professional question is which one, and why.

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SECTION 02

Vivino in Context.

WHAT THE DATASET IS AND HOW IT SKEWS

Vivino is presented as a global consumer wine signal. In practice, it is a collection of ratings with a particular demographic and regional bias, a long tail of low-confidence rows, and a producer field that does not exist as such — only wine names from which producer identity has to be inferred.

That distinction matters. A team should not ask only whether Vivino can answer the question "which Italian producers should Slurpini prioritise?" Vivino can produce a ranking. The better question is whether the ranking it produces resembles the Dutch wine market in the way Slurpini cares about, and whether the analysis surfaces that gap or papers over it.

The right question is not "can this data answer the question?" The right question is what the data gives us that the next-best evidence does not.

What Vivino is good at

Vivino captures large-volume consumer preference at scale. For wines with thousands of reviews — Tenuta Masseto with 4,641, Ornellaia with 581,167, Antinori's Tignanello vintages — the rating is a stable, repeatable proxy for how Dutch consumers actually perceive the bottle. The Bayesian-shrunk weighted rating is the standard tool for this signal: as `n` grows the observed rating dominates, as `n` shrinks the global mean does. The math is uncontroversial.

What Vivino is not

Vivino under-represents the Dutch wine market in two systematic ways. Its user base is Anglocentric and younger. Natural-wine and biodynamic producers are systematically under-counted. Producer names are not a field — they are a string-extraction problem. The pipeline addresses each of those explicitly. A pass-1 alias whitelist resolves the top fifty Italian producers; a pass-2 LLM disambiguation pass on OpenRouter handles the remainder, with a persistent cache so the same dataset never costs twice. A gold-set evaluation against `known_producers_top50.csv` measures the result: **88% recall on exact match, 96% on contains**. Schiopetto and Felsina are the only two misses.



The bias question, quantified

The most important external-validity check in the supercharged track is the bias report. It compares the regional distribution of the cleaned Italian Vivino dataset against the regional shares of Italian wine imports into the Netherlands, drawn from ICE Amsterdam (Italian Trade Agency) annual reports.

The result is honest and uncomfortable. Toscana is over-represented in Vivino by a factor of 1.22. Puglia, by contrast, is under-represented by a factor of 0.61. Abruzzo sits at 0.52 and Campania at 0.55. These are not errors. They are the shape of Vivino's user base bleeding through into the dataset Slurpini would use to decide travel priorities.

Three things follow. First, any recommendation involving Puglia or Abruzzo — both regions where the analysis identifies high-value opportunities — should be marked as *under-sampled*, not low-priority. Second, anyone reading the executive summary should see the bias factor next to the recommendation, not buried in an appendix. Third, the Vivino signal is most reliable in regions Slurpini already knows well, which is the opposite of where it most needs help.

The bare track lists this limitation in four bullets at the end of the recommendation. The supercharged track produces an external CSV (`italian_trade_imports_nl.csv`) and a generated `bias-report.md` table that is regenerated on every run. The point is not to fix the bias. The point is that it should be measured, not hand-waved.

2,986

ITALIAN WINES
AFTER CLEANING

762

DISTINCT PRODUCERS
AFTER DISAMBIGUATION

96%

RECALL ON KNOWN
TOP-50 PRODUCERS

How robust is the top-20?

A ranking is only as defensible as its sensitivity to noise. The supercharged track answers this with a bootstrap of the wine dataset: a thousand resamples with replacement, the producer ranking recomputed in each, the rank of each top-twenty producer recorded across runs. The result is a confidence interval per producer.

Tenuta Masseto appears in the top-ten in **195 out of 200 resamples** in the published run. Dal Forno Romano: 185 out of 200. Biondi-Santi: 160 out of 200. Two producers — Terre di San Vincenzo and Valdicava — appear with extreme p95 ranks (412 and 228 respectively), which means they made the top-ten by accident with too few reviews. That detail does not exist in the bare track. The bare track simply lists them at face value.

The same question applied to the scoring parameter `bayesian_m` gives a sensitivity sweep. At `m` = 200 the top-twenty agrees with itself perfectly. At `m` = 500 the Kendall- τ similarity drops to 0.882. At `m` = 800 it drops to 0.765. The top is stable for the highest-volume producers; the further down the list, the more arbitrary the choice of shrinkage threshold becomes. That, too, is a finding, not a bug.

A producer ranking that does not declare its own stability is not a recommendation. It is a guess presented with confidence.

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COMPOSITE-SCORE
FACTORS PER PRODUCER

3

EXTERNAL BASELINES
USED IN EVALUATION

137

TESTS PASSING
BEFORE ANY SUBMISSION

SECTION 03

From the Field.

USE CASE: SLURPINI PARTNER INTELLIGENCE

Slurpini is a Dutch importer of high-quality Italian wines with an explicit sustainability focus. The company receives more collaboration requests from Italian producers than it can investigate. Each on-site visit costs travel, time and managerial attention. The brief is to add a structured, data-driven layer to that prioritisation problem.

This is a realistic business problem, not a textbook exercise. The challenge is not whether the data can produce a ranking — it can — but whether the ranking will hold up to the constraints of the actual decision: limited budget, sensitive about brand fit, sceptical of black-box scoring, and unforgiving when an importer flies to a region only to discover the data was misleading.

The problem in scope

The technical pipeline is known. Clean and validate the Vivino export (mojibake, tuple-encoded country fields, cross-sheet dedupe). Extract producer identity from wine names — heuristic in the bare track, alias + LLM in the supercharged track. Aggregate per producer with Bayesian shrinkage on review counts. Apply a multi-factor composite score that captures more than just rating. Classify into actionable segments (Hidden Gem, Premium Icon, Commercial Value, Overpriced Risk, Low Confidence Niche) and recommendations (Target, Premium Brand Builder, Value Opportunity, Monitor, Avoid for Now). Quantify bias, declare confidence intervals, surface limitations.

None of these steps require new research. Combining them into a reproducible, defensible pipeline that the buying committee can interrogate is the actual engineering.



The bare route

The bare track is the brief, literally. One Jupyter notebook reads the Vivino export, fixes the obvious data issues, applies a first-word producer heuristic, computes a Bayesian-shrunk weighted rating, ranks the top regions and producer fragments, exports three CSVs and writes a half-page recommendation. The notebook outputs are baked in. The reviewer can read the recommendation in five minutes without running anything. The limitations are surfaced in the same document. "Tenuta San Guido" becomes "Tenuta". Producer extraction is wrong on purpose. The recommendation still arrives at directionally correct findings because the high-volume signal is robust enough that even a crude pipeline can extract it.

The supercharged route

The supercharged track answers the same business question as a data product. Polars and DuckDB instead of pandas. Pandera schema contracts at every stage boundary. Hydra-validated Pydantic configs with config-hash snapshots. A two-pass producer disambiguator with persistent LLM cache. Property-based tests on the scoring math via Hypothesis. Jinja-templated reports whose numbers come from a run log, not from the author. A Vite + React + Recharts dashboard. Fourteen CLI subcommands for operations, comparison, sensitivity sweeps, bootstrap CIs, clustering, bias quantification, anomaly detection, and Firecrawl-driven sustainability lookup.

Comparison

AREA	/BARE ROUTE	/SUPERCHARGED ROUTE
Lines of code	~150, single notebook	~3,500, layered package
Producer extraction	First-word heuristic, ~30% wrong	Alias + LLM disambiguation, 96% recall
Confidence intervals	None	Bootstrap CIs on top-20 ranking
External validity	Four-bullet caveat	Quantified bias vs ICE NL baseline
Reproducibility	"Run all cells"	Config-hash stamped; cantinaiq audit <hash>
Reporting	One markdown file	Eight Jinja-templated artefacts + dashboard
Time to first read	5 minutes	30 minutes
Defensibility under questioning	Limited	Every number traces to a run log

Why both, not one

The temptation is to delete the bare track once the supercharged track is finished. The temptation should be resisted. The bare track is the assignment compliance document. It demonstrates that the brief was answered as briefed, with the limitations the brief is meant to surface. The first-word producer heuristic is deliberately wrong — and the supercharged track exists in part to demonstrate exactly why it is wrong.

The supercharged track is the professional document. It is the version of the analysis I would defend to a buying committee that intends to spend money. The numbers come from a config-hash-stamped run log, traceable end-to-end. The bias is measured and printed at the top of the report, not hand-waved away in a paragraph.

The delta between the two tracks is the argument. Removing either one removes the argument.

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CLI SUBCOMMANDS
IN THE PIPELINE

10

GENERATED REPORTS
PER RUN

1

DASHBOARD SPA OVER
THE SAME EXPORTS

SECTION 04

The Choice.

WHEN THE HEAVYWEIGHT EARNs ITS PLACE AND WHEN IT DOES NOT



When the heavyweight is the right choice

Decisions that will be revisited

If the analysis will be run again — quarterly, annually, on updated data — the cost of reproducibility amortises. Config-hash snapshots mean the seventh run agrees with the first or documents the delta.

Audiences that interrogate methodology

Notebooks do not survive sceptical questioning. A reproducible pipeline does. Every number traces to a parquet, every parquet to a config hash, every hash to a snapshot in version control.

Recommendations that move money

If a buying committee will travel to Italy on the basis of the ranking, the ranking should declare its own confidence. Bootstrap CIs and bias quantification are the difference between defensible and a guess.

Datasets with non-trivial bias

If the data has a known skew — Vivino's Anglo-young demographic — the analysis should measure that skew, not assume it cancels out. The bias report is load-bearing for external validity.

When the heavyweight is the wrong choice

01 / SMALL QUESTIONS

"What's the average rating of wines under €15?"

A one-line pandas query. Building a Polars + DuckDB pipeline for it is theatre, not engineering. The bare track is the right scale.

02 / ONE-OFF ANALYSES

Discovery work that will not be run again

If the result will be screenshot, pasted into a slide and never revisited, reproducibility infrastructure is wasted effort. The cost of building it does not amortise against zero re-runs.

03 / PRE-DECIDED RECOMMENDATIONS

Analyses that are decorative, not decision-supporting

If the recommendation is already made and the analysis is rationalisation, reproducibility makes the rationalisation slightly more defensible without changing the underlying problem. The honest move is to stop producing the analysis.

The deeper question

Most final assignments do not need a reproducible data product. The course brief asks for a notebook and a recommendation, and a notebook and a recommendation is a complete answer. The bare track is the right scale for the assignment in scope.

The supercharged track was built because the same business question — partner selection at a Dutch wine importer — looks materially different when treated as a problem that will be solved more than once. The bias question becomes load-bearing. The bootstrap CIs become the difference between confidence and false confidence. The methodology document becomes the artefact, not the rating.

That is not because the supercharged track is "better" in some absolute sense. It is because the question changed shape. The right tool for the brief-as-stated is the bare track. The right tool for the brief-as-it-would-be-asked-in-industry is the supercharged track. Both are valid. The professional skill is recognising which question is actually being asked.

The better question is not "is more rigour better?" The better question is whether the additional rigour produces a different decision.

For Slurpini, the additional rigour does produce different decisions. The bare track lists "Tenuta", "Marchesi" and "Castello" as top producer fragments — uninterpretable for a buying committee. The supercharged track lists "Tenuta San Guido", "Marchesi Antinori", "Castello di Ama" — directly actionable. The bare track recommends "Expand in Puglia" without flagging that Vivino under-represents Puglia. The supercharged track flags the under-representation factor ($\times 0.61$) next to the recommendation, which changes how the committee should interpret it.

Those are not stylistic improvements. They are decisions changing.

SECTION 05

Closing Observation.

METHODOLOGY CHOICES ARE GOVERNANCE CHOICES



Responsible AI is usually discussed at the model level: fairness, transparency, human oversight, harm reduction. These are important. Responsibility also exists at the methodology level, before any model is fitted.

Which signal is treated as a market signal matters. Which limitations are surfaced matters. Whether confidence intervals are reported matters. Whether the ranking can be reproduced from a snapshot matters. These choices shape what is possible, what is defensible, and what risk is transferred — silently — to the reader of the recommendation.

The Toeslagenaffaire showed what happens when algorithmic systems are treated as neutral facts rather than fallible tools. The Vivino case here is far lower stakes, but the same shape of error is available. A buying committee that reads "Primitivo di Manduria is a value play" without seeing the $\times 0.61$ under-representation factor is the same shape of decision-maker as one who reads "this applicant is high-risk" without seeing the model's confusion matrix on relevant subgroups.

Methodology choices are governance choices.

This document is not an argument against doing the minimum. It is an argument against treating the minimum as automatic. Both tracks answer the brief. They answer different versions of it. The right professional move is not always to build the heavier system. It is to articulate which question is actually being asked and to build the system that answers that question — no more, no less.

But the analysis should still have to earn its trust.

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About the Author.



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This document was written by Vincent Blokker as part of the Final Assignment of the AI Professional programme at the Amsterdam Data Academy. It evaluates the CantinalQ submission from the perspective of when a reproducible data product earns its place against a one-notebook analysis.

The work reflects an applied perspective on data products as a governance question, combining technical evaluation, methodology design and honest accounting of what the data does and does not support. Rigour is treated as a deliberate decision, not a default.

PROGRAMME	MODULE	YEAR
AI Professional	Final Assignment	2026